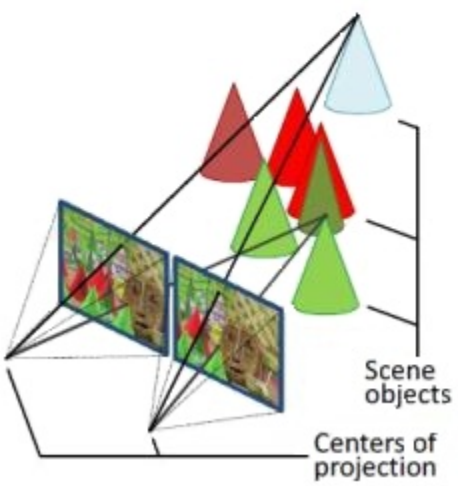
Student Name :______________________

Short Exam related to Calibration Process – Computer Vision 2024

1 h – No document allowed. Al questions are independent (4 questions for 11 points. with 10 points you'll get 20/20)

Question 1 (3 points – 10 mn): Let us consider the following stereo camera system. Let us assume: - that we have seven cones in the scene; - that **we don't know**[1] the 3D coordinates of each point on the top of these cones and - that we know the corresponding points in the left and right views. Let us also assume: - that we know the distance T between the optical centers of the two cameras; - that the focal of the two cameras is the same; - that the scaling parameters along the image directions are the same, and - that there is no skew for each camera. Can we recover the extrinsic and intrinsic calibration parameters of this stereo setup from these seven points only? Justify your answer.



(a) Middlebury stereo image "cones"

(b) left view

Ans:

- Seven unknown parameters (Xi, Yi, Zi, with i varying from 1 to 7)
- Seven pairs of corresponding projected points ($x^l i$, $y^l i$ and ($x^r i$, $y^r i$ with i varying from 1 to 7)
- Assuming that the left camera frame corresponds to the world camera frame, then $x^l = K^l\,[I \mid O]\,\mathbf{X}$ and $x^r = K^r\,[I \mid T]\,\mathbf{X}$. We have therefore only 1 extrinsic parameter (known).
- $K^l = \begin{bmatrix} f\,sx & 0 & o^l x \\ 0 & f\,sy & o^l y \\ 0 & 0 & 1 \end{bmatrix}$ $\quad K^l = \begin{bmatrix} f\,sx & 0 & o^r x \\ 0 & f\,sy & o^r y \\ 0 & 0 & 1 \end{bmatrix}$ We have therefore only 7 intrinsic parameters

[1] In the exam we assumed that these coordinates are known.

- With 4 equations per corresponding points (2 for the left image + 2 for the right image) we have 28 equations (4 * 7 cones) for a total of 21 (3 coord *7 cones) + 7 unknowns to recover, therefore with a SVD approach we can solve such problem.

Question 2 (1 point – 5 mn): Can we state that the pinhole model is linear ? **Hint**: write the corresponding equations using cartesian coordinates, next homogeneous coordinates.

Ans: (see slide 47 of lecture 3a)

as x = f X/Z and y = f Y/Z, we haven't a linear model relying (x, y) to (X, Y, Z)

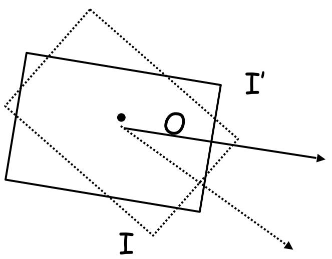
but with homogeneous coordinates we have:

$$Z\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}\begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

Which correspond to a linear model, see more explanations in https://web.stanford.edu/class/cs231a/course_notes/01-camera-models.pdf

Question 3 (3 points – 10 mn): Let us suppose that two images, I and I', are captured by two cameras with internal calibration matrices K and K'. The two cameras have the same camera center and are related by a pure rotation given by matrix R. What is the form of the transformation[2] T relating the two images in terms of K, K' and R? **Hint**: Start from the perspective projection equation, which has the form x = 1/z K [R t] X, where x (3-vector) and X (4-vector) are the image and scene points in homogeneous coordinates, respectively. Assume the two cameras have the same camera center at the origin t1 = t2 = 0.

ANS:



both images I and I' are at the same distance, so t1=t2. If we assume that t1 = t2 = 0 then z = z' and:

x = 1/z K [R1 t1] X
= 1/z K R1 X

x' = 1/z' K' [R2 t2] X
= 1/z K' R2 X

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[2] Such transformation is name an homography.

Considering that $x' = T x$

Then we have $1/z\ K' R2\ X = T\ (1/z\ \ K\ \ R1\ X)$

$$K' R2\ X = T\ K\ \ R1\ X$$

Assuming that there is a R rotation between I and I around the same camera center', then $R2 = R\ R1$ and

$K' R\ R1\ X = T\ K\ R1\ X$

This works with :

$T = K' R\ K^{-1}$

Remind $K^{-1} K = Id$

Question 4 (4 points – 20 mn): Lest us assume that two cameras are looking at the same scene. Their projection matrices are defined as below:

$$P^{(1)} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

and $$P^{(2)} = \begin{bmatrix} 0.8 & 0 & 0.6 & -4 \\ 0 & 1 & 0 & 0 \\ -0.6 & 0 & 0.8 & 3 \end{bmatrix}$$

A 3D world point is project in the first image at the location (2, 0), and in the second image at location (–18, 0) (Each tuple is the (x,y) coordinates). What is the 3D location of this world point?

ANS:

Suppose the world point is $\begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$. From the first camera, we have:

$$\begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \equiv P^{(1)} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \tag{1}$$

$$\Rightarrow \lambda \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} = P^{(1)} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \tag{2}$$

$$\Rightarrow \lambda \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \tag{3}$$

$$\begin{aligned} \Rightarrow 2\lambda &= X \\ 0 &= Y \\ \lambda &= Z \end{aligned} \tag{4}$$

$$\Rightarrow X = 2Z \tag{5}$$

$$Y = 0 \tag{6}$$

Let us now look at the second camerra:

$$\begin{bmatrix} -18 \\ 0 \\ 1 \end{bmatrix} \equiv P^{(2)} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \tag{7}$$

$$\Rightarrow \lambda \begin{bmatrix} -18 \\ 0 \\ 1 \end{bmatrix} = P^{(2)} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \tag{8}$$

$$\begin{aligned} \Rightarrow -18\lambda &= 0.8X + 0.6Z - 4 \\ 0 &= Y \\ \lambda &= -0.6X + 0.8Z + 3 \end{aligned} \tag{9}$$

Substituting $X = 2Z$ in the first and third equations, we get:

$$-18\lambda = 2.2Z - 4 \tag{10}$$

$$\lambda = -0.4Z + 3 \tag{11}$$

$$\Rightarrow -18(-0.4Z + 3) = 2.2Z - 4 \tag{12}$$

$$\Rightarrow 7.2Z - 54 = 2.2Z - 4 \tag{13}$$

$$\Rightarrow 5Z = 50 \tag{14}$$

$$\Rightarrow Z = 10 \tag{15}$$

$$\Rightarrow X = 20 \tag{16}$$

Thus, the point is $(20, 0, 10)$.